

Trigonometric form of Fourier series

<https://www.math24.net/fourier-series-definition-typical-examples/>

Example 22

$$\begin{aligned} y = f(t) &= 0 ; & -4 \leq t \leq 0 \\ y &= 5 ; & 0 \leq t \leq 4 \end{aligned} \quad \text{-----(i)}$$

$$f(t) = f(t+8) \qquad \text{Here, } T = 2L = 8 \qquad \therefore L = 4$$

- a) Sketch the function for 3 cycles:
- b) Find the Fourier series for the function

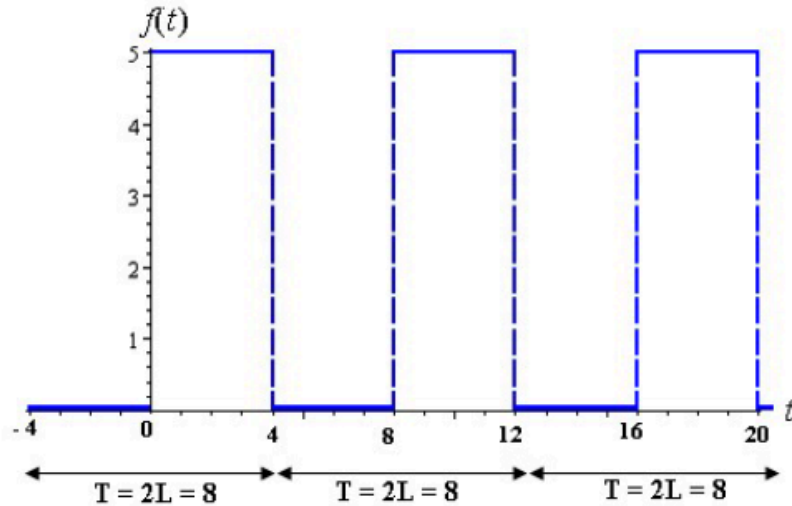


Figure 60: A periodic signal with period $T = 2L = 8$

$$\begin{aligned} a_0 &= \frac{1}{L} \int_{-L}^L f(t) dt \\ &= \frac{1}{4} \int_{-4}^4 f(t) dt \\ &= \frac{1}{4} \int_{-4}^0 f(t) dt + \frac{1}{4} \int_0^4 f(t) dt \\ &= \frac{1}{4} \int_{-4}^0 0 \cdot dt + \frac{1}{4} \int_0^4 5 dt && \text{[From (i)]} \\ &= 0 + \frac{1}{4} [5t]_0^4 && [\int dt = t] \\ &= \frac{1}{4} \times [5 \times 4 - 0] \\ &= \frac{20}{4} = 5 \end{aligned}$$

$$a_n = \frac{1}{L} \int_{-L}^L f(t) \cos \frac{n\pi t}{L} dt$$

$$a_n = \frac{1}{4} \int_{-4}^4 f(t) \cos \frac{n\pi t}{4} dt$$

$$a_n = \frac{1}{4} \int_{-4}^0 f(t) \cos \frac{n\pi t}{4} dt + \frac{1}{4} \int_0^4 f(t) \cos \frac{n\pi t}{4} dt$$

$$a_n = \frac{1}{4} \int_{-4}^0 (0) \cos \frac{n\pi t}{4} dt + \frac{1}{4} \int_0^4 (5) \cos \frac{n\pi t}{4} dt \quad [\text{From (i)}]$$

$$a_n = 0 + \frac{1}{4} \times 5 \int_0^4 \cos \frac{n\pi t}{4} dt$$

$$= \frac{5}{4} \times \frac{1}{\frac{n\pi}{4}} \left[\sin \frac{n\pi t}{4} \right]_0^4 \quad \left[\int \cos mx \, dx = \frac{1}{m} \sin mx \right]$$

$$= \frac{5}{4} \times \frac{4}{n\pi} \left[\sin \frac{n\pi \times 4}{4} - \sin \frac{n\pi \times 0}{4} \right]$$

$$= \frac{5}{n\pi} [\sin n\pi - \sin 0]$$

$$= \frac{5}{n\pi} \sin n\pi$$

$$= 0 \quad [\because \sin \pi = \sin 2\pi = \sin 3\pi = \dots\dots\dots = \sin n\pi = 0 \text{ and } \sin 0 = 0]$$

$$b_n = \frac{1}{L} \int_{-L}^L f(t) \sin \frac{n\pi t}{L} dt$$

$$b_n = \frac{1}{4} \int_{-4}^4 f(t) \sin \frac{n\pi t}{4} dt$$

$$b_n = \frac{1}{4} \int_{-4}^0 f(t) \sin \frac{n\pi t}{4} dt + \frac{1}{4} \int_0^4 f(t) \sin \frac{n\pi t}{4} dt$$

$$b_n = \frac{1}{4} \int_{-4}^0 (0) \sin \frac{n\pi t}{4} dt + \frac{1}{4} \int_0^4 (5) \sin \frac{n\pi t}{4} dt \quad [\text{From (i)}]$$

$$b_n = 0 + \frac{1}{4} \times 5 \int_0^4 \sin \frac{n\pi t}{4} dt$$

$$b_n = \frac{5}{4} \int_0^4 \sin \frac{n\pi t}{4} dt$$

$$= \frac{5}{4} \times \frac{1}{n\pi} \left[-\cos \frac{n\pi t}{4} \right]_0^4 = \frac{-5}{4} \times \frac{4}{n\pi} \left[\cos \frac{n\pi t}{4} \right]_0^4$$

$$\begin{aligned} \int \cos mx \, dx &= \frac{1}{m} \sin mx; \int \sin mx \, dx = \frac{-1}{m} \cos mx \\ &= \frac{-5}{n\pi} \left[\cos \frac{n\pi \times 4}{4} - \cos \frac{n\pi \times 0}{4} \right] \\ &= \frac{-5}{n\pi} [\cos n\pi - \cos 0] \\ &= \frac{-5}{n\pi} [\cos n\pi - 1] \end{aligned}$$

The Fourier series for the above function:

$$\begin{aligned} f(t) &= \frac{a_0}{2} + \sum_{n=1}^{\infty} a_n \cos \frac{n\pi t}{L} + \sum_{n=1}^{\infty} b_n \sin \frac{n\pi t}{L} \\ &= \frac{5}{2} + \sum_{n=1}^{\infty} (0) \cos \frac{n\pi t}{L} + \sum_{n=1}^{\infty} -\frac{5}{n\pi} (\cos(n\pi) - 1) \sin \frac{n\pi t}{L} \\ &= 2.5 - \frac{5}{\pi} \sum_{n=1}^{\infty} \frac{1}{n} (\cos n\pi - 1) \sin \frac{n\pi t}{4} \\ \underbrace{f(t)}_{\text{Complex wave}} &= \underbrace{2.5}_{\text{DC value}} + \underbrace{\left[-\frac{5}{\pi} \sum_{n=1}^{\infty} \frac{1}{n} (\cos n\pi - 1) \sin \frac{n\pi t}{4} \right]}_{\text{AC value}} \text{Answer} \end{aligned}$$

Example 23:

$$\begin{aligned} y = f(t) &= -1; \quad -\pi < t < 0 \\ &= 1; \quad 0 < t < \pi \quad \text{-----(i)} \end{aligned}$$

$$f(t) = f(t + 2\pi) \quad \text{Here, } T = 2L = 2\pi \quad \therefore L = \pi$$

- Sketch the function for 3 cycles:
- Find the Fourier series for the function

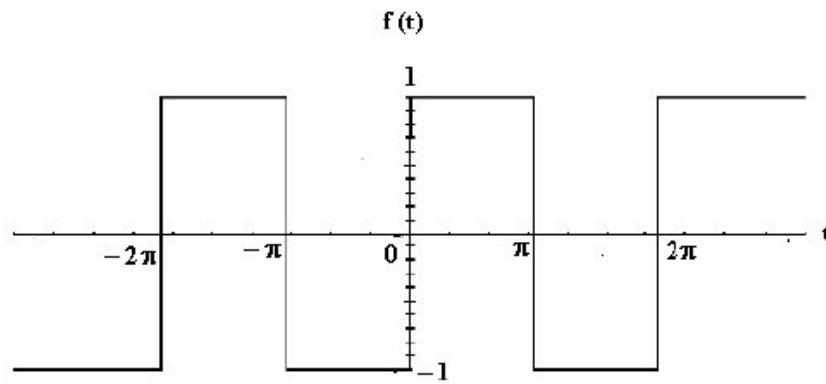


Figure 61: A periodic signal with period $T = 2L = 2\pi$

$$a_0 = \frac{1}{L} \int_{-\pi}^{\pi} f(t) dt$$

$$\begin{aligned} \Rightarrow a_0 &= \frac{1}{\pi} \int_{-\pi}^{\pi} f(t) dt \\ &= \frac{1}{\pi} \int_{-\pi}^0 f(t) dt + \frac{1}{\pi} \int_0^{\pi} f(t) dt \\ &= \frac{1}{\pi} \int_{-\pi}^0 (-1) dt + \frac{1}{\pi} \int_0^{\pi} (1) dt && \text{[From (i)]} \\ &= \frac{-1}{\pi} [t]_{-\pi}^0 + \frac{1}{\pi} [t]_0^{\pi} \\ &= \frac{-1}{\pi} [0 - (-\pi)] + \frac{1}{\pi} [\pi - 0] \\ &= -\frac{1}{\pi} [\pi] + \frac{1}{\pi} [\pi] \\ &= -1 + 1 \\ &= 0 \end{aligned}$$

$$a_n = \frac{1}{L} \int_{-\pi}^{\pi} f(t) \cos \frac{n\pi t}{L} dt$$

$$\begin{aligned} a_n &= \frac{1}{\pi} \int_{-\pi}^{\pi} f(t) \cos \frac{n\pi t}{\pi} dt \\ &= \frac{1}{\pi} \int_{-\pi}^0 f(t) \cos \frac{n\pi t}{\pi} dt + \frac{1}{\pi} \int_0^{\pi} f(t) \cos \frac{n\pi t}{\pi} dt \end{aligned}$$

$$\begin{aligned}
&= \frac{1}{\pi} \int_{-\pi}^0 (-1) \cos nt \, dt + \frac{1}{\pi} \int_0^{\pi} (1) \cos nt \, dt \quad [\text{From (i)}] \\
&= -\frac{1}{\pi} \int_{-\pi}^0 \cos nt \, dt + \frac{1}{\pi} \int_0^{\pi} \cos nt \, dt \\
&= -\frac{1}{\pi} \times \frac{1}{n} [\sin nt]_{-\pi}^0 + \frac{1}{\pi} \times \frac{1}{n} [\sin nt]_0^{\pi} \quad \left[\int \cos mx \, dx = \frac{1}{m} \sin mx \right] \\
&= -\frac{1}{\pi n} [\sin 0 - \sin(-n\pi)] + \frac{1}{\pi n} [\sin n\pi - 0] \\
&= -\frac{1}{\pi n} [0 - \sin(-n\pi)] + \frac{1}{\pi n} [\sin n\pi - 0] \\
&= -\frac{1}{\pi n} [0 + \sin(n\pi)] + \frac{1}{\pi n} [\sin n\pi - 0] \quad [\because \sin(-\theta) = -\sin \theta] \\
&= -\frac{1}{\pi n} [0 + 0] + \frac{1}{\pi n} [0 - 0] \quad [\because \sin n\pi = 0 \text{ for } n = 1, 2, 3, \dots] \\
&= 0
\end{aligned}$$

$$b_n = \frac{1}{L} \int_{-\pi}^{\pi} f(t) \sin \frac{n\pi t}{L} dt$$

$$\begin{aligned}
b_n &= \frac{1}{\pi} \int_{-\pi}^{\pi} f(t) \sin \frac{n\pi t}{\pi} dt \\
&= \frac{1}{\pi} \int_{-\pi}^0 f(t) \sin \frac{n\pi t}{\pi} dt + \frac{1}{\pi} \int_0^{\pi} f(t) \sin \frac{n\pi t}{\pi} dt \\
&= \frac{1}{\pi} \int_{-\pi}^0 f(t) \sin nt \, dt + \frac{1}{\pi} \int_0^{\pi} f(t) \sin nt \, dt \\
&= \frac{1}{\pi} \int_{-\pi}^0 (-1) \sin nt \, dt + \frac{1}{\pi} \int_0^{\pi} (1) \sin nt \, dt \quad [\text{From (i)}] \\
&= -\frac{1}{\pi} \int_{-\pi}^0 \sin nt \, dt + \frac{1}{\pi} \int_0^{\pi} \sin nt \, dt \\
&= -\frac{1}{\pi} \times \frac{1}{n} [-\cos nt]_{-\pi}^0 + \frac{1}{\pi} \times \frac{1}{n} [-\cos nt]_0^{\pi} \quad \left[\because \int \sin mx \, dx = -\frac{1}{m} \cos mx \right] \\
&= +\frac{1}{\pi n} [\cos nt]_{-\pi}^0 + \frac{1}{\pi n} [-\cos nt]_0^{\pi} \\
&= \frac{1}{\pi n} [\cos 0 - \cos(-n\pi)] - \frac{1}{\pi n} [\cos n\pi - \cos 0] \\
&= \frac{1}{\pi n} [1 - \cos n\pi] - \frac{1}{\pi n} [\cos n\pi - 1] \quad [\because \cos(-\theta) = \cos \theta] \\
&= \frac{1}{\pi n} - \frac{1}{\pi n} \cos n\pi - \frac{1}{\pi n} \cos n\pi + \frac{1}{\pi n}
\end{aligned}$$

$$= \frac{2}{\pi n} - \frac{2}{\pi n} \cos n\pi$$

$$= \frac{2}{\pi n} (1 - \cos n\pi)$$

The Fourier series for the above function:

$$\therefore f(t) = \frac{a_0}{2} + \sum_{n=1}^{\infty} a_n \cos \frac{n\pi t}{L} + \sum_{n=1}^{\infty} b_n \sin \frac{n\pi t}{L}$$

$$\therefore f(t) = \frac{a_0}{2} + \sum_{n=1}^{\infty} a_n \cos \frac{n\pi t}{\pi} + \sum_{n=1}^{\infty} b_n \sin \frac{n\pi t}{\pi}$$

$$\therefore f(t) = \frac{a_0}{2} + \sum_{n=1}^{\infty} a_n \cos nt + \sum_{n=1}^{\infty} b_n \sin nt$$

$$\therefore f(t) = 0 + \sum_{n=1}^{\infty} 0 \cdot \cos nt + \sum_{n=1}^{\infty} \frac{2}{n\pi} (1 - \cos n\pi) \sin nt$$

$$\underbrace{f(t)}_{\text{Complex wave}} = \underbrace{0}_{\text{DC value}} + \underbrace{\sum_{n=1}^{\infty} \frac{2}{n\pi} (1 - \cos n\pi) \sin nt}_{\text{AC value}}$$

Example 24:

$$y = f(t) = -t; \quad -\pi \leq t \leq 0$$

$$y = t; \quad 0 \leq t \leq \pi \quad \text{-----(i)}$$

$$f(t) = f(t + 2\pi) \quad \text{Here, } T = 2L = 2\pi \quad \therefore L = \pi$$

- Sketch the function for 3 cycles:
- Find the Fourier series for the function

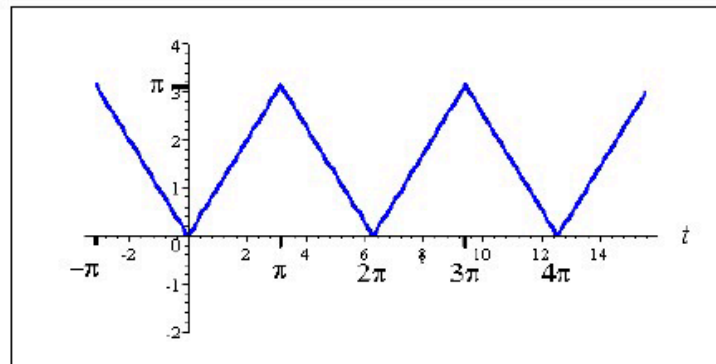


Figure 62: A periodic signal with period $T = 2L = 2\pi$

$$a_0 = \frac{1}{L} \int_{-L}^L f(t) dt$$

$$a_0 = \frac{1}{\pi} \int_{-\pi}^{\pi} f(t) dt$$

$$\begin{aligned}
&= \frac{1}{\pi} \int_{-\pi}^0 f(t) dt + \frac{1}{\pi} \int_0^{\pi} f(t) dt \\
&= \frac{1}{\pi} \int_{-\pi}^0 (-t) dt + \frac{1}{\pi} \int_0^{\pi} t dt \quad [\text{From (i)}] \\
&= -\frac{1}{\pi} \left[\frac{t^2}{2} \right]_{-\pi}^0 + \frac{1}{\pi} \left[\frac{t^2}{2} \right]_0^{\pi} \quad \left[\because \int x^n dx = \frac{x^{n+1}}{n+1} + c ; n \neq -1 \right] \\
&= -\frac{1}{\pi} \left[0 - \frac{(-\pi)^2}{2} \right] + \frac{1}{\pi} \left[\frac{\pi^2}{2} - 0 \right] \\
&= -\frac{1}{\pi} \left[-\frac{\pi^2}{2} \right] + \frac{1}{\pi} \left[\frac{\pi^2}{2} \right] \\
&= \frac{1}{\pi} \times \frac{\pi^2}{2} + \frac{1}{\pi} \times \frac{\pi^2}{2} \\
&= \frac{\pi}{2} + \frac{\pi}{2} = \frac{2\pi}{2} \\
&= \pi
\end{aligned}$$

$$\begin{aligned}
a_n &= \frac{1}{L} \int_{-L}^L f(t) \cos \frac{n\pi t}{L} dt \\
a_n &= \frac{1}{\pi} \int_{-\pi}^{\pi} f(t) \cos \frac{n\pi t}{\pi} dt \\
a_n &= \frac{1}{\pi} \int_{-\pi}^{\pi} f(t) \cos(nt) dt \\
&= \frac{1}{\pi} \int_{-\pi}^0 f(t) \cos(nt) dt + \frac{1}{\pi} \int_0^{\pi} f(t) \cos(nt) dt \\
&= \frac{1}{\pi} \int_{-\pi}^0 (-t) \cos(nt) dt + \frac{1}{\pi} \int_0^{\pi} t \cos(nt) dt \quad [\text{From (i)}] \\
&= -\frac{1}{\pi} \int_{-\pi}^0 t \cos(nt) dt + \frac{1}{\pi} \int_0^{\pi} t \cos(nt) dt \text{-----(ii)}
\end{aligned}$$

Now, Let, $I = \int t \cos(nt) dt$

$$\begin{aligned}
&= t \int \cos(nt) dt - \int \left[\frac{d}{dt} (t) \int \cos(nt) dt \right] dt \quad \left[\because \int u v dx = u \int v dx - \int \left\{ \frac{d}{dx} (u) \int v dx \right\} dx \right] \\
&= t \frac{\sin(nt)}{n} - \int 1 \cdot \frac{\sin(nt)}{n} dt \quad \left[\int \cos mx dx = \frac{1}{m} \sin mx \right] \\
&= \frac{t}{n} \sin(nt) - \frac{1}{n} \int \sin(nt) dt \\
&= \frac{t}{n} \sin(nt) - \frac{1}{n} \frac{(-\cos(nt))}{n} \quad \left[\because \int \sin mx dx = -\frac{1}{m} \cos mx \right]
\end{aligned}$$

$$= \frac{t}{n} \sin(nt) + \frac{1}{n^2} \cos(nt) \text{-----(iii)}$$

Putting the value of (iii) in (ii)

$$\begin{aligned} a_n &= \frac{1}{\pi} \int_{-\pi}^{\pi} f(t) \cos(nt) dt = -\frac{1}{\pi} \int_{-\pi}^0 t \cos(nt) dt + \frac{1}{\pi} \int_0^{\pi} t \cos(nt) dt \\ &= -\frac{1}{\pi} \left[\frac{t}{n} \sin(nt) + \frac{1}{n^2} \cos(nt) \right]_{-\pi}^0 + \frac{1}{\pi} \left[\frac{t}{n} \sin(nt) + \frac{1}{n^2} \cos(nt) \right]_0^{\pi} \\ &= -\frac{1}{\pi} \left[\frac{0}{n} \sin(0) + \frac{1}{n^2} \cos(0) - \frac{(-\pi)}{n} \sin(-n\pi) - \frac{1}{n^2} \cos(-n\pi) \right] + \frac{1}{\pi} \left[\frac{\pi}{n} \sin(n\pi) + \frac{1}{n^2} \cos(n\pi) \right. \\ &\quad \left. - \frac{0}{n} \sin(0) - \frac{1}{n^2} \cos(0) \right] \\ &= -\frac{1}{\pi} \left[0 + \frac{1}{n^2} \times 1 + \frac{\pi}{n} \sin(-n\pi) - \frac{1}{n^2} \cos(-n\pi) \right] + \frac{1}{\pi} \left[\frac{\pi}{n} \times 0 + \frac{1}{n^2} \cos(n\pi) - 0 - \frac{1}{n^2} \times 1 \right] \\ &= -\frac{1}{\pi} \left[\frac{1}{n^2} - \frac{\pi}{n} \sin(n\pi) - \frac{1}{n^2} \cos(n\pi) \right] + \frac{1}{\pi} \left[\frac{1}{n^2} \cos(n\pi) - \frac{1}{n^2} \right] \\ &\quad [\sin(-\theta) = -\sin \theta; \cos(-\theta) = \cos \theta; \sin n\pi = 0 \text{ for } n = 1, 2, 3, \dots] \\ &= -\frac{1}{\pi} \left[\frac{1}{n^2} - \frac{\pi}{n} \times 0 - \frac{1}{n^2} \cos(n\pi) \right] + \frac{1}{\pi} \left[\frac{1}{n^2} \cos(n\pi) - \frac{1}{n^2} \right] \\ &= \frac{-1}{\pi n^2} + \frac{1}{\pi n^2} \cos(n\pi) + \frac{1}{\pi n^2} \cos(n\pi) - \frac{1}{\pi n^2} \\ &= \frac{2}{\pi n^2} \cos n\pi - \frac{2}{\pi n^2} \\ &= \frac{2}{\pi n^2} (\cos n\pi - 1) \quad \quad \quad \left[\text{Taking Common } \frac{2}{\pi n^2} \right] \end{aligned}$$

$$b_n = \frac{1}{L} \int_{-\pi}^{\pi} f(t) \sin \frac{n\pi t}{L} dt$$

$$b_n = \frac{1}{\pi} \int_{-\pi}^{\pi} f(t) \sin \frac{n\pi t}{\pi} dt$$

$$\begin{aligned} b_n &= \frac{1}{\pi} \int_{-\pi}^{\pi} f(t) \sin(nt) dt \\ &= \frac{1}{\pi} \int_{-\pi}^0 f(t) \sin(nt) dt + \frac{1}{\pi} \int_0^{\pi} f(t) \sin(nt) dt \\ &= \frac{1}{\pi} \int_{-\pi}^0 (-t) \sin(nt) dt + \frac{1}{\pi} \int_0^{\pi} t \sin(nt) dt \\ &= -\frac{1}{\pi} \int_{-\pi}^0 t \sin(nt) dt + \frac{1}{\pi} \int_0^{\pi} t \sin(nt) dt \text{-----(iv)} \end{aligned}$$

Now $\int t \sin(nt) dt$

$$\begin{aligned}
&= t \int \sin(nt) dt - \int \left\{ \frac{d}{dt}(t) \int \sin(nt) dt \right\} dt \quad [\because \int u v dx = u \int v dx - \int \left\{ \frac{d}{dx}(u) \int v dx \right\} dx] \\
&= t \left(\frac{-\cos(nt)}{n} \right) - \int 1 \cdot \frac{-\cos(nt)}{n} dt \\
&= -\frac{t}{n} \cos(nt) + \frac{1}{n} \int \cos(nt) dt \\
&= -\frac{t}{n} \cos(nt) + \frac{1}{n} \frac{\sin(nt)}{n} \quad \left[\int \cos mx dx = \frac{1}{m} \sin mx \right] \\
&= -\frac{t}{n} \cos(nt) + \frac{1}{n^2} \sin(nt) \text{------(v)}
\end{aligned}$$

Putting the value of (v) in (iv)

$$\begin{aligned}
b_n &= \frac{1}{\pi} \int_{-\pi}^{\pi} f(t) \sin(nt) dt = -\frac{1}{\pi} \int_{-\pi}^0 t \sin(nt) dt + \frac{1}{\pi} \int_0^{\pi} t \sin(nt) dt \\
&= -\frac{1}{\pi} \left[-\frac{t}{n} \cos(nt) + \frac{1}{n^2} \sin(nt) \right]_{-\pi}^0 + \frac{1}{\pi} \left[-\frac{t}{n} \cos(nt) + \frac{1}{n^2} \sin(nt) \right]_0^{\pi} \\
&= -\frac{1}{\pi} \left[\frac{-0}{n} \cos(0) + \frac{1}{n^2} \sin(0) - \frac{-(-\pi)}{n} \cos(-n\pi) - \frac{1}{n^2} \sin(-n\pi) \right] + \frac{1}{\pi} \left[-\frac{\pi}{n} \cos(n\pi) \right. \\
&\quad \left. + \frac{1}{n^2} \sin(n\pi) - \frac{- (0)}{n} \cos(0) - \frac{1}{n^2} \sin(0) \right] \\
&= -\frac{1}{\pi} \left[0 + 0 - \frac{\pi}{n} \cos(n\pi) + \frac{1}{n^2} \sin(n\pi) \right] + \frac{1}{\pi} \left[-\frac{\pi}{n} \cos(n\pi) + \frac{1}{n^2} \sin(n\pi) + 0 - 0 \right] \\
&= -\frac{1}{\pi} \left[-\frac{\pi}{n} \cos(n\pi) + \frac{1}{n^2} \times 0 \right] + \frac{1}{\pi} \left[-\frac{\pi}{n} \cos(n\pi) + \frac{1}{n^2} \times 0 \right] \\
&\quad [\sin(-\theta) = -\sin \theta ; \cos(-\theta) = \cos \theta ; \sin n\pi = 0 \text{ for } n = 1, 2, 3, \dots] \\
&= \frac{\pi}{\pi n} \cos(n\pi) - \frac{\pi}{\pi n} \cos(n\pi) \\
&= 0
\end{aligned}$$

The Fourier series for the above function is:

$$\begin{aligned}
f(t) &= \frac{a_0}{2} + \sum_{n=1}^{\infty} a_n \cos\left(\frac{n\pi}{L}t\right) + \sum_{n=1}^{\infty} b_n \sin\left(\frac{n\pi}{L}t\right) \\
&= \frac{\pi}{2} + \sum_{n=1}^{\infty} \frac{2}{n^2 \pi} (\cos n\pi - 1) \cos\left(\frac{n\pi}{L}t\right) + 0 \\
&= \frac{\pi}{2} + \sum_{n=1}^{\infty} \frac{2}{n^2 \pi} (\cos n\pi - 1) \cos\left(\frac{n\pi}{L}t\right) \\
\underbrace{f(t)}_{\text{Complex wave}} &= \underbrace{\frac{\pi}{2}}_{\text{DC value}} + \underbrace{\sum_{n=1}^{\infty} \frac{2}{n^2 \pi} (\cos n\pi - 1) \cos nt}_{\text{AC value}}
\end{aligned}$$

Example 25

$$y = f(t) = t ; -\pi \leq t \leq \pi \text{-----(i)}$$

$$\text{Here, } T = 2L = 2\pi \quad \therefore L = \pi$$

- Sketch the function for 3 cycles:
- Find the Fourier series for the function

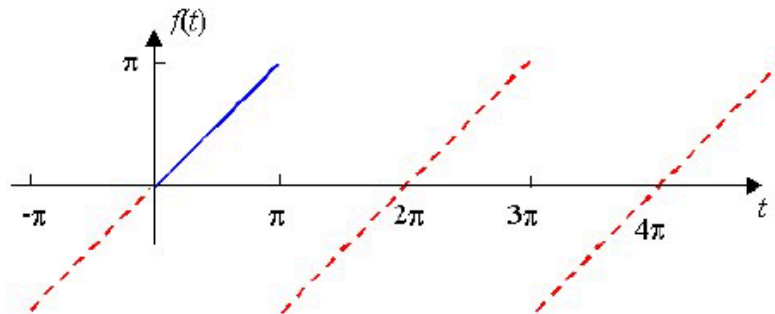


Figure 63: A periodic signal with period $T = 2L = 2\pi$

$$a_0 = \frac{1}{L} \int_{-L}^L f(t) dt$$

$$a_0 = \frac{1}{\pi} \int_{-\pi}^{\pi} f(t) dt$$

$$a_0 = \frac{1}{\pi} \int_{-\pi}^{\pi} t dt \quad [\text{From (i)}]$$

$$= \frac{1}{\pi} \left[\frac{t^2}{2} \right]_{-\pi}^{\pi}$$

$$= \frac{1}{\pi} \left[\frac{\pi^2}{2} - \frac{(-\pi)^2}{2} \right]$$

$$= \frac{1}{\pi} \left[\frac{\pi^2}{2} - \frac{\pi^2}{2} \right]$$

$$= 0$$

$$a_n = \frac{1}{\pi} \int_{-\pi}^{\pi} f(t) \cos \frac{n\pi t}{L} dt$$

$$= \frac{1}{\pi} \int_{-\pi}^{\pi} f(t) \cos \frac{n\pi t}{\pi} dt$$

$$= \frac{1}{\pi} \int_{-\pi}^{\pi} f(t) \cos n t dt$$

$$= \frac{1}{\pi} \int_{-\pi}^{\pi} t \cos n t dt \quad [\text{From (i)}]$$

Now $\int t \cos nt dt$

$$\begin{aligned}
 &= t \int \cos nt dt - \int \left\{ \frac{d}{dt}(t) \int \cos nt dt \right\} dt \quad [\because \int u v dx = u \int v dx - \int \left\{ \frac{d}{dx}(u) \int v dx \right\} dx] \\
 &= \frac{t}{n} \times \sin nt - \int 1 \cdot \frac{\sin nt}{n} dt \\
 &= \frac{t}{n} \sin nt - \frac{1}{n} \cdot \frac{1}{n} \cdot (-\cos nt) \\
 &= \frac{t}{n} \sin nt + \frac{1}{n^2} \cos(nt) \text{-----(ii)}
 \end{aligned}$$

$$\begin{aligned}
 \therefore a_n &= \frac{1}{\pi} \int_{-\pi}^{\pi} t \cos nt dt \\
 &= \frac{1}{\pi} \left[\frac{t}{n} \sin nt + \frac{1}{n^2} \cos nt \right]_{-\pi}^{\pi} \quad [\text{From (ii)}] \\
 &= \frac{1}{\pi} \left[\frac{\pi}{n} \sin n\pi + \frac{1}{n^2} \cos n\pi - \left\{ \frac{-\pi}{n} \sin(-n\pi) + \frac{1}{n^2} \cos(-n\pi) \right\} \right] \\
 &= \frac{1}{\pi} \left[\frac{\pi}{n} \sin n\pi + \frac{1}{n^2} \cos n\pi + \frac{\pi}{n} \sin(-n\pi) - \frac{1}{n^2} \cos(-n\pi) \right] \\
 &= \frac{1}{\pi} \left[\frac{\pi}{n} \sin n\pi + \frac{1}{n^2} \cos n\pi + \frac{\pi}{n} (-) \sin(n\pi) - \frac{1}{n^2} \cos(n\pi) \right] \\
 &\quad [\because \sin(-\theta) = -\sin \theta, \cos(-\theta) = \cos \theta] \\
 &= \frac{1}{\pi} \left[\frac{\pi}{n} \cancel{\sin n\pi} + \frac{1}{n^2} \cos n\pi - \frac{\pi}{n} \cancel{\sin n\pi} - \frac{1}{n^2} \cos n\pi \right] \\
 &= 0
 \end{aligned}$$

$$\begin{aligned}
 b_n &= \frac{1}{\pi} \int_{-\pi}^{\pi} f(t) \cdot \sin \frac{n\pi t}{L} dt \\
 &= \frac{1}{\pi} \int_{-\pi}^{\pi} f(t) \sin \frac{n\pi t}{\pi} dt \\
 &= \frac{1}{\pi} \int_{-\pi}^{\pi} f(t) \sin nt dt \\
 &= \frac{1}{\pi} \int_{-\pi}^{\pi} t \sin nt dt \quad [\text{From (i)}]
 \end{aligned}$$

Now, $\int t \sin nt dt$

$$\begin{aligned}
 &= t \int \sin nt dt - \int \left\{ \frac{d}{dt}(t) \int \sin nt dt \right\} dt \quad [\because \int u v dx = u \int v dx - \int \left\{ \frac{d}{dx}(u) \int v dx \right\} dx] \\
 &= t \cdot \frac{1}{n} \cdot (-\cos nt) - \int 1 \cdot \frac{1}{n} \cdot (-\cos nt) dt
 \end{aligned}$$

$$\begin{aligned}
&= -\frac{t}{n} \cos nt + \frac{1}{n} \int \cos nt dt \\
&= -\frac{t}{n} \cos nt + \frac{1}{n^2} \sin nt \text{-----(iii) } \left[\int \cos mx dx = \frac{1}{m} \sin mx \right]
\end{aligned}$$

$$\begin{aligned}
\therefore b_n &= \frac{1}{\pi} \int_{-\pi}^{\pi} t \sin nt dt \\
&= \frac{1}{\pi} \left[-\frac{t}{n} \cos nt + \frac{1}{n^2} \sin nt \right]_{-\pi}^{\pi} \quad \text{[From (iii)]} \\
&= \frac{1}{\pi} \left[-\frac{\pi}{n} \cos n\pi + \frac{1}{n^2} \sin n\pi - \left\{ -\frac{-\pi}{n} \cos(-n\pi) + \frac{1}{n^2} \sin(-n\pi) \right\} \right] \\
&= \frac{1}{\pi} \left[-\frac{\pi}{n} \cos n\pi + \frac{1}{n^2} \sin n\pi - \left\{ \frac{\pi}{n} \cos(-n\pi) + \frac{1}{n^2} \sin(-n\pi) \right\} \right] \\
&= \frac{1}{\pi} \left[-\frac{\pi}{n} \cos n\pi + \frac{1}{n^2} \sin n\pi - \frac{\pi}{n} \cos(-n\pi) - \frac{1}{n^2} \sin(-n\pi) \right] \\
&= \frac{1}{\pi} \left[-\frac{\pi}{n} \cos n\pi + \frac{1}{n^2} \sin n\pi - \frac{\pi}{n} \cos(-n\pi) - \frac{1}{n^2} (-) \sin(n\pi) \right] \\
&\quad [\because \sin(-\theta) = -\sin \theta ; \cos(-\theta) = \cos \theta] \\
&= \frac{1}{\pi} \left[-\frac{\pi}{n} \cos n\pi + \frac{1}{n^2} \sin n\pi - \frac{\pi}{n} \cos n\pi + \frac{1}{n^2} \sin n\pi \right] \\
&= \frac{1}{\pi} \left[-\frac{\pi}{n} \cos n\pi - \frac{\pi}{n} \cos n\pi \right] \quad [\because \sin n\pi = 0, \text{ for } n = 1, 2, 3, \dots] \\
&= -\frac{1}{\pi} \times \frac{2\pi}{n} \cos n\pi \\
&= -\frac{2}{n} \cos n\pi
\end{aligned}$$

The Fourier series for the above function is:

$$\begin{aligned}
\therefore f(t) &= \frac{a_0}{2} + \sum_{n=1}^{\infty} a_n \cos nt + \sum_{n=1}^{\infty} b_n \sin nt \\
&= 0 + 0 + \sum_{n=1}^{\infty} b_n \sin nt \\
&= \sum_{n=1}^{\infty} -\frac{2}{n} \cos n\pi \cdot \sin nt \\
\underbrace{f(t)}_{\text{Complex wave}} &= \underbrace{0}_{\text{DC value}} + \underbrace{-2 \sum_{n=1}^{\infty} \frac{1}{n} \cos n\pi \cdot \sin nt}_{\text{AC value}}
\end{aligned}$$

Complex form of Fourier series: Ex-37

Example 37: Find the complex form of the Fourier Series expansion of the periodic function $f(t)$ and find the trigonometric form of the Fourier Series expansion of the periodic function $f(t)$ is given by:

$$f(t) = \cos \frac{1}{2}t \quad ; -\pi < t < \pi \quad [T = 2L = 2\pi]$$

Solution:

The complex form of the Fourier series expansion of a periodic function $f(t)$, of period T , is:

$$f(t) = \sum_{n=-\infty}^{\infty} c_n e^{in\omega t} \text{-----(A)}$$

Where,

$$c_0 = \frac{1}{T} \int_{-L}^L f(t) dt$$

$$c_n = \frac{1}{T} \int_{-L}^L f(t) e^{-in\omega t} dt$$

$$c_n^* = c_{-n} = \frac{1}{T} \int_{-L}^L f(t) e^{in\omega t} dt$$

$$c_n = \frac{1}{T} \int_{-L}^L f(t) e^{-in\omega t} dt$$

$$c_n = \frac{1}{2\pi} \int_{-\pi}^{\pi} f(t) e^{-in\omega t} dt$$

$$= \frac{1}{2\pi} \int_{-\pi}^{\pi} f(t) e^{-in \frac{2\pi}{T} t} dt$$

$$= \frac{1}{2\pi} \int_{-\pi}^{\pi} f(t) e^{-in \frac{2\pi}{2\pi} t} dt \quad [T = 2\pi]$$

$$= \frac{1}{2\pi} \int_{-\pi}^{\pi} f(t) e^{-int} dt$$

$$= \frac{1}{2\pi} \int_{-\pi}^{\pi} \cos \frac{1}{2}t e^{-int} dt$$

$$= \frac{1}{2\pi} \int_{-\pi}^{\pi} \left\{ \frac{1}{2} (e^{i(\frac{1}{2}t)} + e^{-i(\frac{1}{2}t)}) \right\} e^{-int} dt \quad [\because \cos x = \frac{1}{2}(e^{ix} + e^{-ix})]$$

$$\begin{aligned}
&= \frac{1}{4\pi} \int_{-\pi}^{\pi} \{ (e^{i(\frac{1}{2}t)} + e^{-i(\frac{1}{2}t)}) e^{-int} \} dt \\
&= \frac{1}{4\pi} \int_{-\pi}^{\pi} \{ (e^{i(\frac{1}{2}t)} \cdot e^{-int} + e^{-i(\frac{1}{2}t)} \cdot e^{-int}) \} dt \\
&= \frac{1}{4\pi} \int_{-\pi}^{\pi} \{ (e^{i(\frac{1}{2}t-nt)} + e^{-i(\frac{1}{2}t+nt)}) \} dt \\
&= \frac{1}{4\pi} \int_{-\pi}^{\pi} \{ (e^{-i(n-\frac{1}{2})t} + e^{-i(n+\frac{1}{2})t}) \} dt \\
&= \frac{1}{4\pi} \left[\frac{e^{-i(n-\frac{1}{2})t}}{-i(n-\frac{1}{2})} + \frac{e^{-i(n+\frac{1}{2})t}}{-i(n+\frac{1}{2})} \right]_{-\pi}^{\pi} \\
&= \frac{1}{4\pi} \left[\frac{2e^{-i(2n-1)t}}{-i(2n-1)} - \frac{2e^{-i(2n+1)t}}{i(2n+1)} \right]_{-\pi}^{\pi} \\
&= \frac{1}{4\pi} \left[\frac{-2e^{-int} \cdot e^{\frac{it}{2}}}{i(2n-1)} - \frac{2e^{-int} \cdot e^{-\frac{it}{2}}}{i(2n+1)} \right]_{-\pi}^{\pi} \\
&= \frac{-2}{4\pi} \left[\frac{e^{-int} \cdot e^{\frac{it}{2}}}{i(2n-1)} + \frac{e^{-int} \cdot e^{-\frac{it}{2}}}{i(2n+1)} \right]_{-\pi}^{\pi} \\
&= \frac{-2}{4\pi} \left[\frac{e^{-in\pi} \cdot e^{\frac{i\pi}{2}}}{i(2n-1)} + \frac{e^{-in\pi} \cdot e^{-\frac{i\pi}{2}}}{i(2n+1)} - \frac{e^{-in(-\pi)} \cdot e^{\frac{-i\pi}{2}}}{i(2n-1)} - \frac{e^{-in(-\pi)} \cdot e^{-\frac{i(-\pi)}{2}}}{i(2n+1)} \right] \\
&= \frac{-2}{4\pi} \left[\frac{e^{-in\pi} \cdot e^{\frac{i\pi}{2}}}{i(2n-1)} + \frac{e^{-in\pi} \cdot e^{-\frac{i\pi}{2}}}{i(2n+1)} - \frac{e^{in\pi} \cdot e^{\frac{-i\pi}{2}}}{i(2n-1)} - \frac{e^{in\pi} \cdot e^{\frac{i\pi}{2}}}{i(2n+1)} \right] \text{------(B)}
\end{aligned}$$

We have,

$$e^{ix} = \cos x + i \sin x$$

$$\therefore e^{i\frac{\pi}{2}} = \cos \frac{\pi}{2} + i \sin \frac{\pi}{2} = 0 + i.1 = i \text{-----} (*)$$

$$\text{And, } e^{-ix} = \cos x - i \sin x$$

$$\therefore e^{-i\frac{\pi}{2}} = \cos \frac{\pi}{2} - i \sin \frac{\pi}{2} = 0 - i.1 = -i \text{-----} (*)$$

$$e^{in\pi} = \cos n\pi + i \sin n\pi = (-1)^n + i.0 = (-1)^n \text{-----} (*)$$

Since,

$$\text{if } n=1, \cos n\pi = \cos \pi = -1$$

$$\text{if } n=2, \cos n\pi = \cos 2\pi = 1$$

$$\text{if } n=3, \cos n\pi = \cos 3\pi = -1$$

Hence we can write,

$$\cos n\pi = (-1)^n [n = 1, 2, 3, \dots]$$

Again,

$$e^{-in\pi} = \cos n\pi - i \sin n\pi = (-1)^n - i.0 = (-1)^n \text{-----} (*)$$

Putting these values in (B),

$$c_n = \frac{-2}{4\pi} \left[\frac{e^{-in\pi} \cdot e^{\frac{i\pi}{2}}}{i(2n-1)} + \frac{e^{-in\pi} \cdot e^{-\frac{i\pi}{2}}}{i(2n+1)} - \frac{e^{in\pi} \cdot e^{\frac{i\pi}{2}}}{i(2n-1)} - \frac{e^{in\pi} \cdot e^{-\frac{i\pi}{2}}}{i(2n+1)} \right]$$

$$c_n = \frac{-2}{4\pi} \left[\frac{e^{-in\pi} \cdot i}{i(2n-1)} + \frac{e^{-in\pi} \cdot (-i)}{i(2n+1)} - \frac{e^{in\pi} \cdot (-i)}{i(2n-1)} - \frac{e^{in\pi} \cdot (i)}{i(2n+1)} \right]$$

$$c_n = \frac{-2}{4\pi} \left[\frac{e^{-in\pi} \cdot i}{i(2n-1)} - \frac{e^{-in\pi} \cdot (i)}{i(2n+1)} + \frac{e^{in\pi} \cdot (i)}{i(2n-1)} - \frac{e^{in\pi} \cdot (i)}{i(2n+1)} \right]$$

$$c_n = \frac{-2}{4\pi} \left[\frac{(-1)^n \cdot i}{i(2n-1)} - \frac{(-1)^n \cdot (i)}{i(2n+1)} + \frac{(-1)^n \cdot (i)}{i(2n-1)} - \frac{(-1)^n \cdot (i)}{i(2n+1)} \right]$$

$$c_n = \frac{-2}{4\pi} \left[\frac{(-1)^n}{(2n-1)} - \frac{(-1)^n}{(2n+1)} + \frac{(-1)^n}{(2n-1)} - \frac{(-1)^n}{(2n+1)} \right]$$

$$c_n = \frac{-2}{4\pi} \left[\frac{1}{(2n-1)} - \frac{1}{(2n+1)} + \frac{1}{(2n-1)} - \frac{1}{(2n+1)} \right] (-1)^n$$

$$c_n = \frac{-2}{4\pi} \left[\frac{2}{(2n-1)} - \frac{2}{(2n+1)} \right] (-1)^n$$

$$c_n = \frac{-4}{4\pi} \left[\frac{1}{(2n-1)} - \frac{1}{(2n+1)} \right] (-1)^n$$

$$c_n = \frac{1}{\pi} \left[\frac{1}{(2n+1)} - \frac{1}{(2n-1)} \right] (-1)^n$$

$$c_n = \frac{1}{\pi} \left[\frac{2n-1-(2n+1)}{(2n+1)(2n-1)} \right] (-1)^n$$

$$c_n = \frac{1}{\pi} \left[\frac{-2}{(2n+1)(2n-1)} \right] (-1)^n$$

$$c_n = \frac{-2}{\pi} \left[\frac{1}{(4n^2-1)} \right] (-1)^n$$

$$c_n = \frac{2}{\pi} \left[\frac{1}{(4n^2-1)} \right] (-1)^n (-1)$$

$$c_n = \frac{2}{\pi} \left[\frac{1}{(4n^2 - 1)} \right] (-1)^{n+1}$$

∴ From (A), the complex form of the Fourier series expansion of a periodic function $f(t)$, of period T , is:

$$\begin{aligned} f(t) &= \sum_{n=-\infty}^{\infty} c_n e^{in\omega t} \\ &= \sum_{n=-\infty}^{\infty} \left\{ \frac{2}{\pi} \left[\frac{1}{(4n^2 - 1)} \right] (-1)^{n+1} \right\} e^{in\omega t} \\ &= \sum_{n=-\infty}^{\infty} \left\{ \frac{2}{\pi} \left[\frac{1}{(4n^2 - 1)} \right] (-1)^{n+1} \right\} e^{int} \left[\omega = \frac{2\pi}{T} = \frac{2\pi}{2\pi} = 1 \right] \text{ Answer} \end{aligned}$$

Line spectrum

Example 33:

A Fourier series is given below. List frequencies and plot the one-sided amplitude spectrum

$$\underbrace{x(t)}_{\text{Complex wave}} = \underbrace{12}_{\text{DC value}} + \underbrace{9 \cos(2\pi \times 10t + \frac{\pi}{3}) + 6 \cos(2\pi \times 20t - \frac{\pi}{6}) + 4 \cos(2\pi \times 30t + \frac{\pi}{4})}_{\text{AC value}}$$

$$[\because \omega = 2\pi f]$$

Here DC value is 12; Amplitudes are 9, 6, and 4 with respect to frequencies 10, 20,

30 respectively where phases are $\frac{\pi}{3}, -\frac{\pi}{6}, \frac{\pi}{4}$

The frequencies are 0 (dc), 10 Hz, 20 Hz, 30 Hz

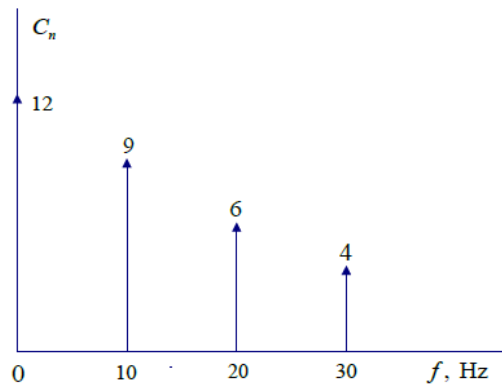


Figure 76

Example 34:

A plot showing each of the harmonic amplitudes in the wave is called the line spectrum.

Plot the line spectrum (discrete frequency spectra) for the Fourier series:

$$f(t) = \frac{\pi}{2} + \sum_{n=1}^{\infty} \frac{1}{2n-1} \cos nt + \sum_{n=1}^{\infty} \frac{(-1)^n}{2n} \sin nt \text{(i)}$$

This series has an interesting graph for the above function **f(t)**

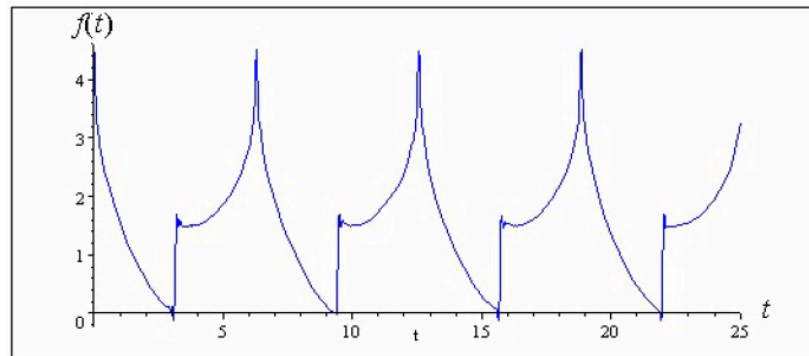


Figure 82

We have the Fourier series is

$$\underbrace{f(t)}_{\text{Complex wave}} = \underbrace{\frac{a_0}{2}}_{\text{DC value}} + \underbrace{\sum_{n=1}^{\infty} a_n \cos(n\omega t) + \sum_{n=1}^{\infty} b_n \sin(n\omega t)}_{\text{AC value}} \text{-----(ii)}$$

Where the variables are:

- $\frac{a_0}{2} \rightarrow$ The *average* (DC) value of the signal:
- The a_n holds the amplitudes of the cosine wave
- The b_n holds the amplitudes of the sine wave
- $n \rightarrow$ The *harmonic number*: 1=fundamental, 2=2nd harmonic, etc
- $n\omega \rightarrow$ frequency

Comparing (i) & (ii) that

$$\frac{a_0}{2} = \frac{\pi}{2}, a_n = \frac{1}{2n-1}, b_n = \frac{(-1)^n}{2n}, n\omega = n$$

From problem 10, we have

$$\text{If } a \cos \theta + b \sin \theta = R \sin(\theta + \alpha)$$

Here,

$a \rightarrow$ Amplitude of the cosine wave

$b \rightarrow$ Amplitude of the sine wave

$R \rightarrow$ Amplitude of the new signal

$\alpha \rightarrow$ Phase Shift

$$\text{Then } R = \sqrt{a^2 + b^2}$$

Accordingly, for the equation (i), we can write, $R_n = C_n = \sqrt{a_n^2 + b_n^2}$

$a_n = \frac{1}{2n-1}$	$b_n = \frac{(-1)^n}{2n}$	$C_n = \sqrt{a_n^2 + b_n^2}$
$a_1 = 1$	$b_1 = -\frac{1}{2}$	$C_1 = \sqrt{1^2 + \left(-\frac{1}{2}\right)^2} = 1.118$
$a_2 = \frac{1}{3}$	$b_2 = \frac{1}{4}$	$C_2 = \sqrt{\left(\frac{1}{3}\right)^2 + \left(\frac{1}{4}\right)^2} = 0.4167$
$a_3 = \frac{1}{5}$	$b_3 = -\frac{1}{6}$	$C_3 = \sqrt{\left(\frac{1}{5}\right)^2 + \left(-\frac{1}{6}\right)^2} = 0.260$
$a_4 = \frac{1}{7}$	$b_4 = \frac{1}{8}$	$C_4 = \sqrt{\left(\frac{1}{7}\right)^2 + \left(\frac{1}{8}\right)^2} = 0.190$

Here, from (i) & (ii)

Here, $n\omega = n$

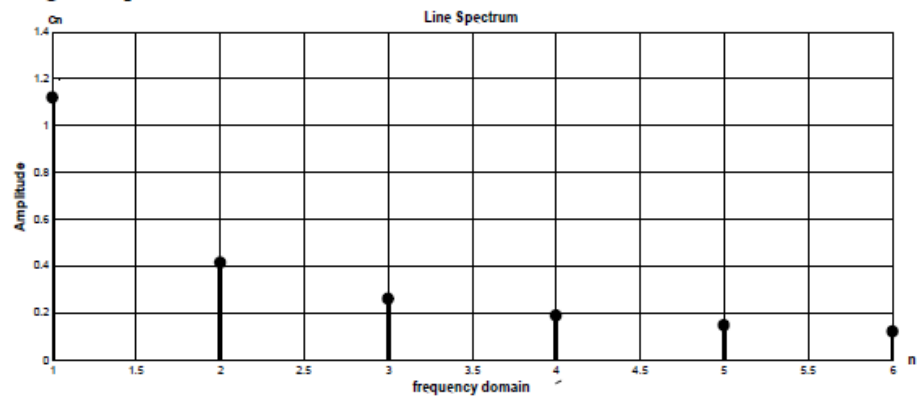
For $n = 1$; Fundamental Frequency = 1st Harmonic = $\omega = 1$

For $n = 2$; 2nd Harmonic = $2\omega = 2$

For $n = 3$; 3rd Harmonic = $3\omega = 3$

For $n = 4$; 4th Harmonic = $4\omega = 4$

The resulting line spectrum is:



Home Task:

Plot the line spectrum (discrete frequency spectra) for the Fourier series (Answer of example 22):

$$\underbrace{f(t)}_{\text{Complex wave}} = \underbrace{2.5}_{\text{DC value}} - \underbrace{\frac{5}{\pi} \sum_{n=1}^{\infty} \frac{1}{n} (\cos n\pi - 1)}_{\text{Amplitude}} \sin \underbrace{\frac{n\pi}{4}}_{\text{Frequency}} t \text{-----(i)}$$

We have the Fourier series is

$$\underbrace{f(t)}_{\text{Complex wave}} = \underbrace{\frac{a_0}{2}}_{\text{DC value}} + \underbrace{\sum_{n=1}^{\infty} a_n \cos(n\omega t) + \sum_{n=1}^{\infty} b_n \sin(n\omega t)}_{\text{AC value}} \text{-----(ii)}$$

Where the variables are:

- $\frac{a_0}{2} \rightarrow$ The *average* (DC) value of the signal:
- The a_n holds the amplitudes of the cosine wave
- The b_n holds the amplitudes of the sine wave
- $n \rightarrow$ The *harmonic number*: 1=fundamental, 2=2nd harmonic, etc
- $n\omega \rightarrow$ frequency

Comparing (i) & (ii) that

$$\frac{a_0}{2} = 2.5, a_n = 0; b_n = -\frac{5}{\pi} \frac{1}{n} (\cos n\pi - 1), n\omega = \frac{n\pi}{4}$$

$$\text{Now, using } \Rightarrow R = \sqrt{a^2 + b^2}$$

$$\text{Let } R_n = C_n = \sqrt{a_n^2 + b_n^2}$$

$a_n = 0$	$b_n = -\frac{5}{\pi} \frac{1}{n} (\cos n\pi - 1)$	$R_n = C_n = \sqrt{a_n^2 + b_n^2}$
$a_1 = 0$	$b_1 = -\frac{5}{\pi} \frac{1}{1} (\cos \pi - 1)$ $b_1 = -\frac{5}{\pi} \frac{1}{1} (-1 - 1)$ $b_1 = \frac{10}{\pi}$	$R_1 = C_1 = \sqrt{a_1^2 + b_1^2}$ $R_1 = C_1 = \sqrt{0^2 + \left(\frac{10}{\pi}\right)^2}$ $R_1 = C_1 = \frac{10}{\pi}$ $R_1 = C_1 = \frac{10}{3.14} = 3.18$
$a_2 = ?$	$b_2 = ?$	$R_2 = C_2 = \sqrt{a_2^2 + b_2^2}$
$a_3 = ?$	$b_3 = ?$	$R_3 = C_3 = \sqrt{a_3^2 + b_3^2}$
$a_4 = ?$	$b_4 = ?$	$R_4 = C_4 = \sqrt{a_4^2 + b_4^2}$
-----	-----	-----

Here, $n\omega = \frac{n\pi}{4}$

For $n = 1$; Fundamental Frequency = 1st Harmonic = $\omega = \frac{\pi}{4} = 0.785$

For $n = 2$; 2nd Harmonic = $2\omega = \frac{2\pi}{4} = 1.570$

For $n = 3$; 3rd Harmonic = $3\omega = \frac{3\pi}{4} = 2.356$

For $n = 4$; 4th Harmonic = $4\omega = \frac{4\pi}{4} = 3.1416$

Plot line spectrum (Do the rest yourself)

Convolution Sum, Convolution Integral

Convolution is a mathematical tool for combining two signals to produce a third signal.

The following steps are to be taken

- i. Folding
- ii. Shifting
- iii. Multiplication
- iv. Summation

1st times:

- i. Folding
- ii. Multiplication
- iii. Summation

2nd times and more

- i. Shifting
- ii. Multiplication
- iii. Summation

Example 102:

Evaluate the convolution sums of $y[n] = x[n] * h[n]$

Where,

$x[n] = 1, n = 0$ and $h[n] = 2, n = 0$; n represents the time index
 $3, n = 1$ $1, n = 1$

MATLAB

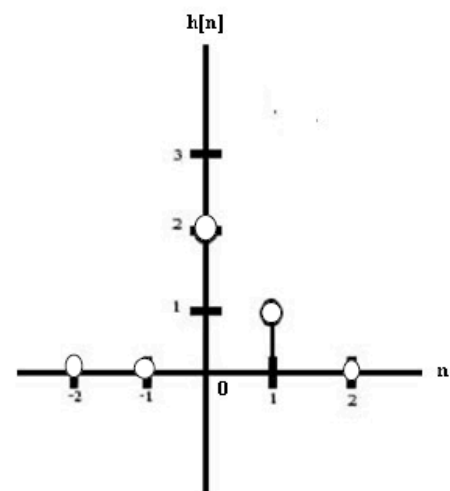
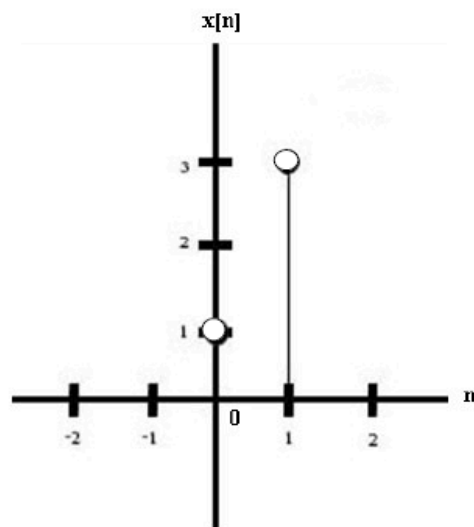
`x=[1 3]`

`h=[2 1]`

`y=conv(x,h)`

`y= [2 7 3]`

Solution:



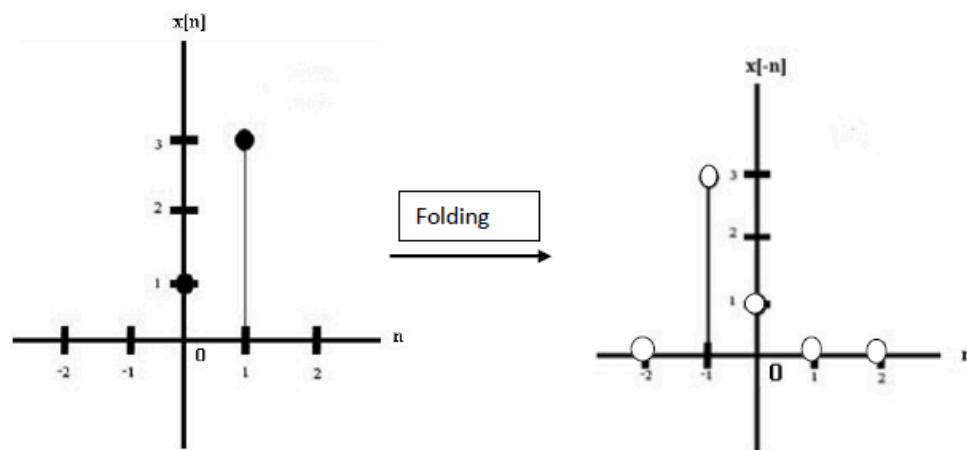
1st time:

(i). Folding:

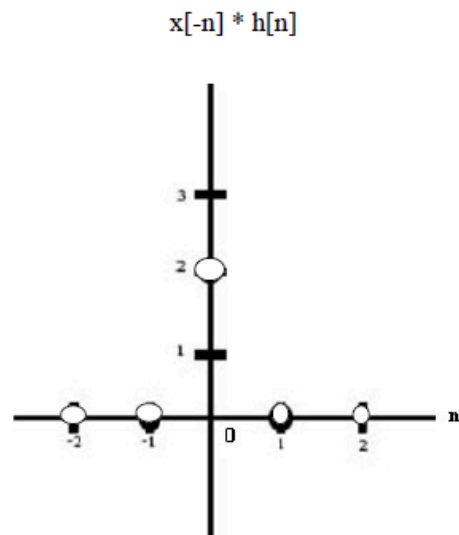
$$x[n] = x[-n]$$

$$\text{i.e. } x[0] = x[0]$$

$$x[1] = x[-1]$$



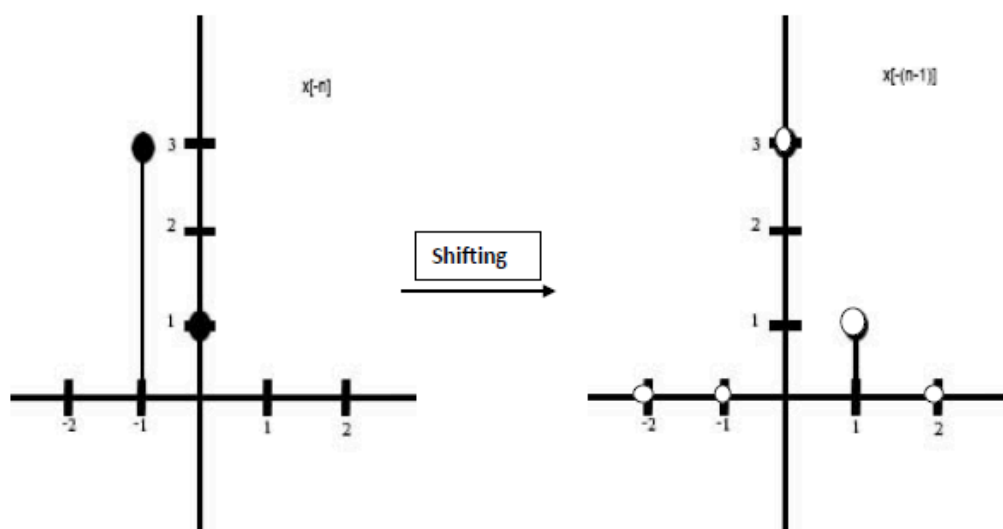
(ii). Multiplication:



(iii). Summation: $y[0] = 0 + 2 + 0 = 2$

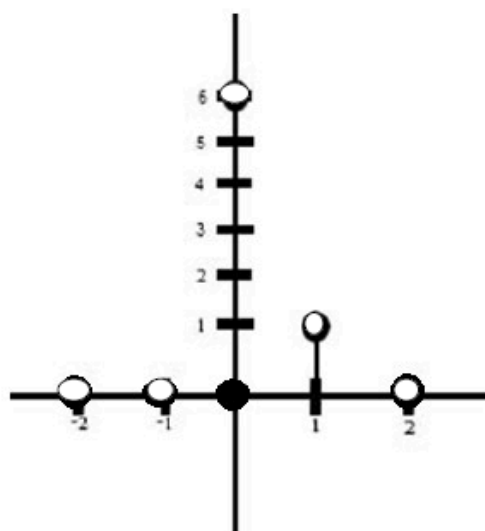
2nd time:

(i). Shifting:



(ii) Multiplication:

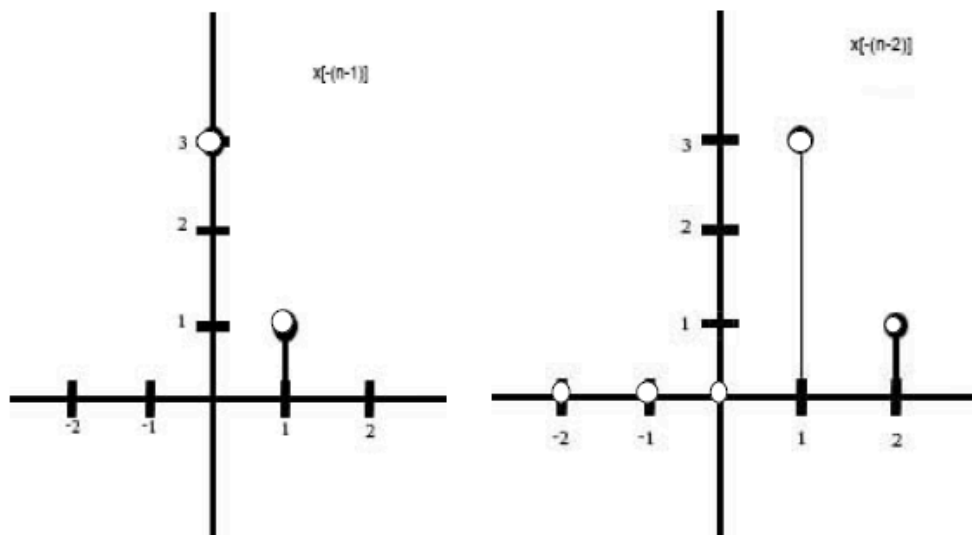
$$x[-(n-1)] * h[n]$$



(iii). Summation: $y[1] = 0 + 6 + 1 + 0 = 7$

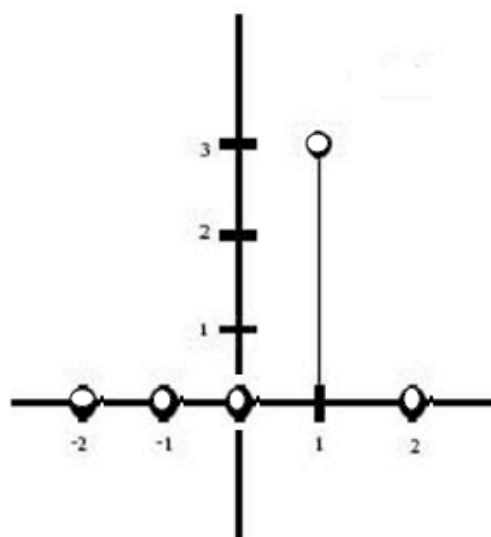
3rd time:

(i). Shifting:



(ii). Multiplication:

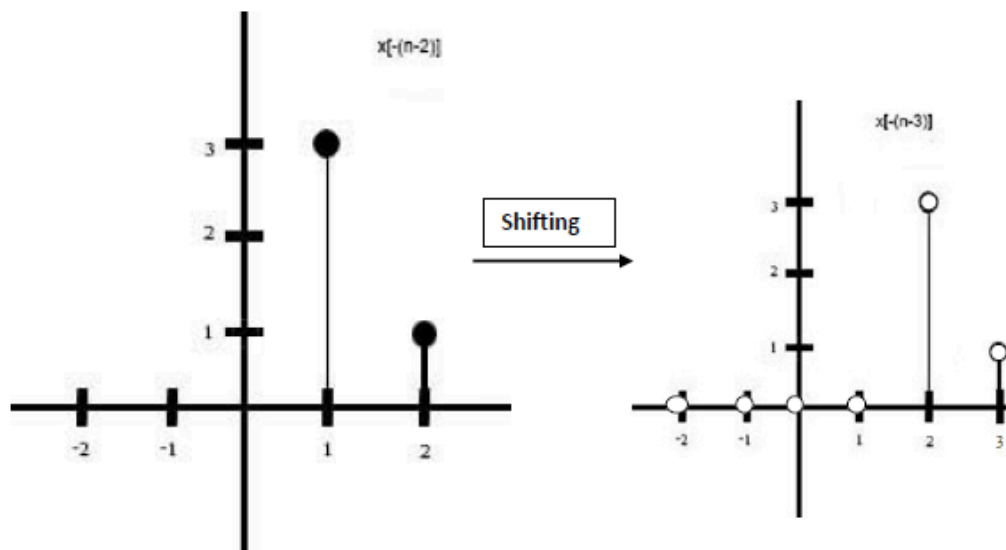
$$x[-(n-2)] * h[n]$$



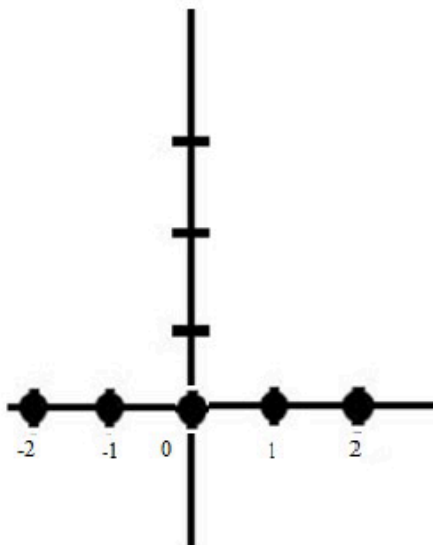
(iii). Summation: $y[2] = 0 + 0 + 3 + 0 = 3$

4th time:

(i). Shifting:



(ii). Multiplication: $x[-(n-3)] * h[n]$



iii) Summation: $y[3] = 0+0+0+0=0$

Finally we get,

$$\therefore y[0] = 2$$

$$y[1] = 7$$

$$y[2] = 3$$

$$\therefore \text{Convolution Sum: } y[n] = [2 \ 7 \ 3]$$

Problem 36: Convolution Integral

- i. Folding
- ii. Shifting
- iii. Finding Area

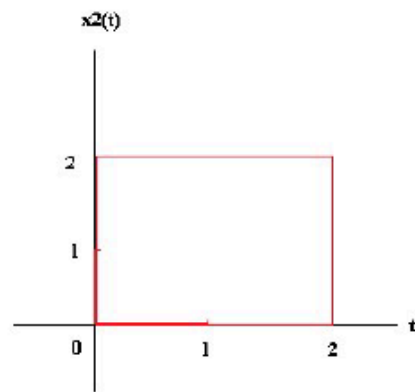
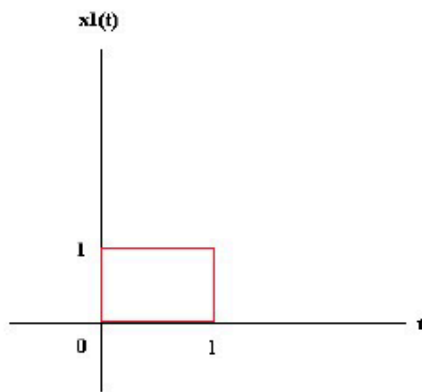
1st time:

- i. **Folding**
- ii. Finding Area

2nd time and more

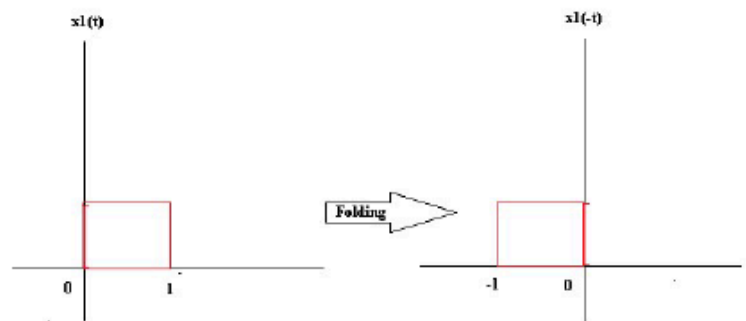
- i. Shifting
- ii. Finding Area

Example 107: Find convolution integral of $x_1(t) * x_2(t)$

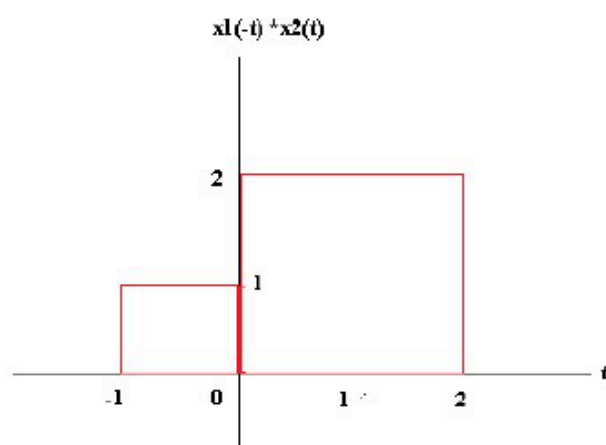


1st time:

- i. Folding



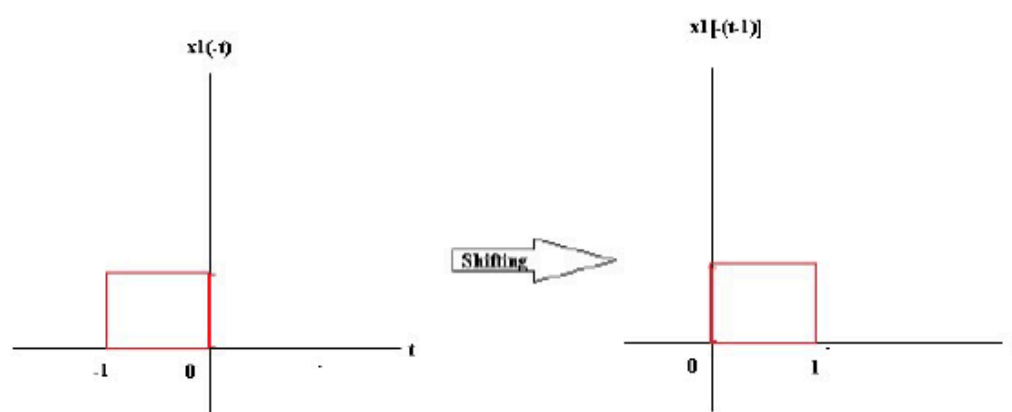
ii. Finding Area



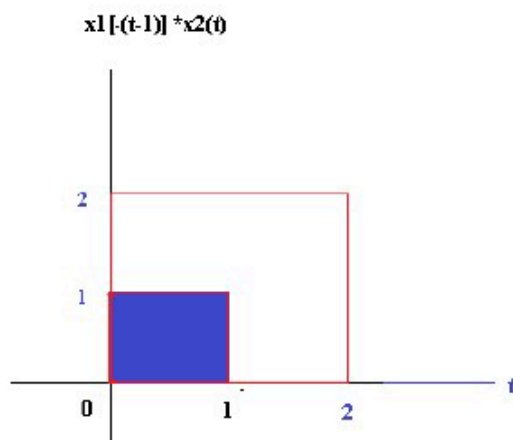
Finding Area: No overlapped area; $x[0] = 0$

2nd time:

i. Shifting

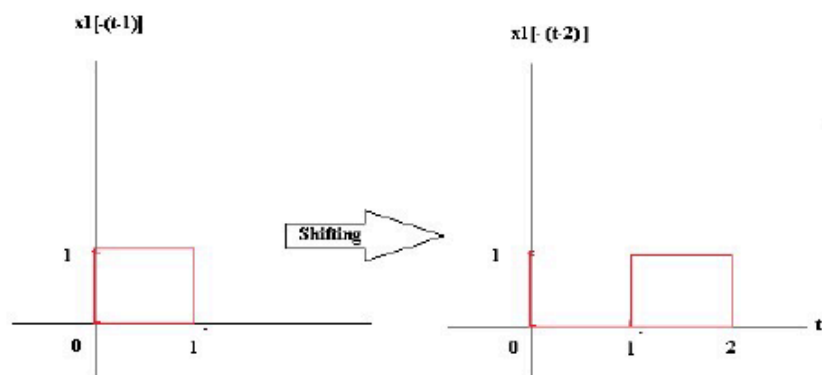


ii. Finding Area:

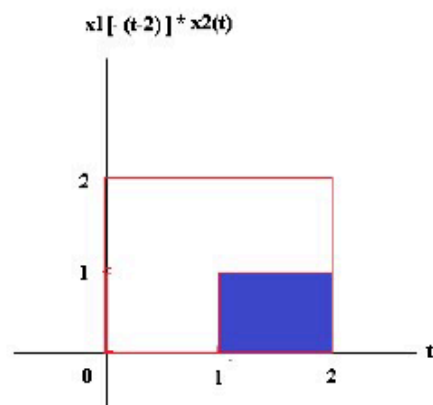


Area: $x[1] = 1 \times 1 = 1$

iii. Shifting

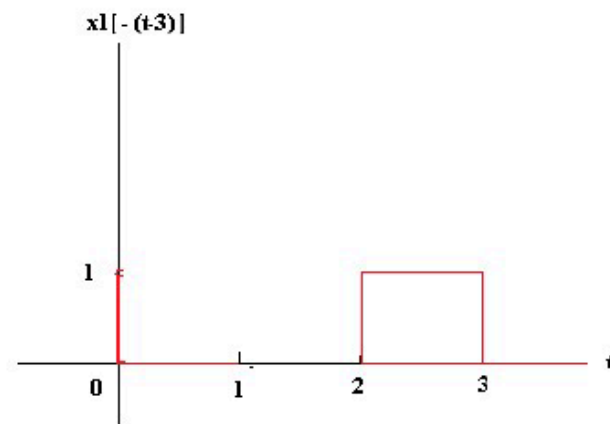


iv. Finding area

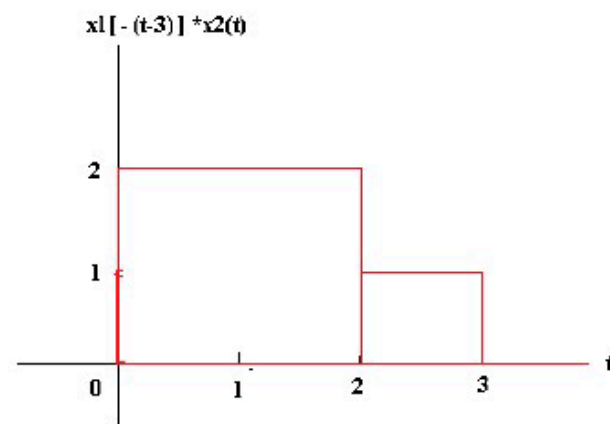


Area: $x[2] = 1 \times 1 = 1$

v. Shifting:



vi. Finding Area:



No overlapped: $x[3] = 0$

Hence the convolution integral is

